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If a step is a question, answer the question after you solve it! -> print("This is the answer")

In [1]:
```
import pandas as pd
```

Import all the files

In [2]:
```
visits = pd.read_csv('Data/visits-1.csv',
                     parse_dates=[1])

cart = pd.read_csv('Data/cart-1.csv',
                   parse_dates=[1])

checkout = pd.read_csv('Data/checkout-1.csv',
                       parse_dates=[1])

purchase = pd.read_csv('Data/purchase-1.csv',
                       parse_dates=[1])
```

Step 1: Inspect the DataFrames using `print` and `head`

In [3]:
```
print(visits.head())
print(cart.head())
print(checkout.head())
print(purchase.head())
```

```
                                 user_id          visit_time
0  943647ef-3682-4750-a2e1-918ba6f16188 2017-04-07 15:14:00
1  0c3a3dd0-fb64-4eac-bf84-ba069ce409f2 2017-01-26 14:24:00
2  6e0b2d60-4027-4d9a-babd-0e7d40859fb1 2017-08-20 08:23:00
3  6879527e-c5a6-4d14-b2da-50b85212b0ab 2017-11-04 18:15:00
4  a84327ff-5daa-4ba1-b789-d5b4caf81e96 2017-02-27 11:25:00
                                 user_id           cart_time
0  2be90e7c-9cca-44e0-bcc5-124b945ff168 2017-11-07 20:45:00
1  4397f73f-1da3-4ab3-91af-762792e25973 2017-05-27 01:35:00
2  a9db3d4b-0a0a-4398-a55a-ebb2c7adf663 2017-03-04 10:38:00
3  b594862a-36c5-47d5-b818-6e9512b939b3 2017-09-27 08:22:00
4  a68a16e2-94f0-4ce8-8ce3-784af0bbb974 2017-07-26 15:48:00
                                 user_id       checkout_time
0  d33bdc47-4afa-45bc-b4e4-dbe948e34c0d 2017-06-25 09:29:00
1  4ac186f0-9954-4fea-8a27-c081e428e34e 2017-04-07 20:11:00
2  3c9c78a7-124a-4b77-8d2e-e1926e011e7d 2017-07-13 11:38:00
3  89fe330a-8966-4756-8f7c-3bdbcd47279a 2017-04-20 16:15:00
4  3ccdaf69-2d30-40de-b083-51372881aedd 2017-01-08 20:52:00
                                 user_id       purchase_time
0  4b44ace4-2721-47a0-b24b-15fbfa2abf85 2017-05-11 04:25:00
1  02e684ae-a448-408f-a9ff-dcb4a5c99aac 2017-09-05 08:45:00
2  4b4bc391-749e-4b90-ab8f-4f6e3c84d6dc 2017-11-20 20:49:00
3  a5dbb25f-3c36-4103-9030-9f7c6241cd8d 2017-01-22 15:18:00
4  46a3186d-7f5a-4ab9-87af-84d05bfd4867 2017-06-11 11:32:00
```

Step 2: Left merging visits and cart

In [4]:
```
visits_cart = visits.merge(cart, how='left')
```

Step 3: How long is `visits_cart`?

In [5]:
```
print(len(visits_cart))
```

```
2000
```

Step 4: How many timestamps are null for `cart_time`?

In [6]:
```
null_cart_time = visits_cart.cart_time.isnull().sum()
print(null_cart_time)
```

```
1652
```

Step 5: What percentage only visited?

In [7]:
```
visited_not_cart = null_cart_time / len(visits_cart)
print(f'{visited_not_cart:.2%}')
```

```
82.60%
```

Step 6: What percentage placed a t-shirt in their cart but did not checkout?

In [8]:
```
cart.merge(checkout, how='left').head()
```

Out[8]:

	user_id	cart_time	checkout_time
0	2be90e7c-9cca-44e0-bcc5-124b945ff168	2017-11-07 20:45:00	2017-11-07 21:14:00
1	4397f73f-1da3-4ab3-91af-762792e25973	2017-05-27 01:35:00	NaT
2	a9db3d4b-0a0a-4398-a55a-ebb2c7adf663	2017-03-04 10:38:00	2017-03-04 11:04:00
3	b594862a-36c5-47d5-b818-6e9512b939b3	2017-09-27 08:22:00	2017-09-27 08:26:00
4	a68a16e2-94f0-4ce8-8ce3-784af0bbb974	2017-07-26 15:48:00	NaT

In [9]:
```
cart_checkout = cart.merge(checkout, how='left')

no_checkout_time = cart_checkout.checkout_time.isnull().sum()
no_checkout_time_pct = no_checkout_time / len(cart_checkout)

print(f"{no_checkout_time_pct:.2%} of shoppers placed a t-shirt in their car
```

```
35.06% of shoppers placed a t-shirt in their cart but did not checkout.
```

Step 7: Merge it all together

In [10]:
```
all_data = visits.merge(cart, how='left').merge(checkout, how='left').merge(
```

Step 8: % of users who got to checkout but did not purchase

In [11]:
```
checkout_users = all_data[all_data.checkout_time.notnull()]
checkout_not_purchase = checkout_users.purchase_time.isnull().sum()
checkout_not_purchase_pct = checkout_not_purchase / len(all_data)

print(f"{checkout_not_purchase_pct:.2%} of shoppers checked out but did not
```

```
3.89% of shoppers checked out but did not purchase a t-shirt.
```

Step 9: check each part of the funnel, let's print all 3 of them again

In [12]:
```
print("{} percent of users who visited the page did not add a t-shirt to the
print("{} percent of users who added a t-shirt to their cart did not checkou
print("{} percent of users who made it to checkout  did not purchase a shirt
```

```
82.6 percent of users who visited the page did not add a t-shirt to their ca
rt
35.06 percent of users who added a t-shirt to their cart did not checkout
3.89 percent of users who made it to checkout  did not purchase a shirt
```

The weakest part of the funnel is clearly getting a person who visited the site to add a tshirt to their cart. Once they've added a t-shirt to their cart it is fairly likely they end up purchasing it. A suggestion could be to make the add-to-cart button more prominent on the front page.

Step 10: adding new column

In [13]:
```
all_data['time_to_purchase'] = all_data.purchase_time - all_data.visit_time
```

Step 11: examine the results

In [14]:
```
print(all_data.time_to_purchase.dropna())
```

```
1      0 days 00:44:00
14     0 days 00:38:00
48     0 days 00:41:00
49     0 days 01:00:00
65     0 days 01:00:00
             ...
2083   0 days 00:29:00
2093   0 days 00:46:00
2097   0 days 00:32:00
2098   0 days 00:28:00
2099   0 days 00:11:00
Name: time_to_purchase, Length: 252, dtype: timedelta64[ns]
```

Step 12: what is the average time to purchase?

In [15]:
```
print(all_data.time_to_purchase.mean())
```

```
0 days 00:43:12.380952380
```